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EMPIRICAL RESEARCH

The roles of contextual elements in post-merger common platform development: an empirical investigation

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Abstract
With congeneric mergers, which involve firms with interrelated but not identical business lines that develop diverse products and services, a major challenge for organizations is the development of a common platform that fulfills similar business functions across multiple divisions. Through a field study of a post-merger common platform development initiative, we develop a framework that highlights how environmental and organizational contexts shape the process of common platform development (CPD). Our study provides an account of how the focal organization transitioned to a platform-based approach by achieving a balance between stable and flexible aspects of the common platform through negotiations among the divisions acquired through mergers and acquisitions. These negotiations were enabled through various boundary-spanning activities and the guided successive enrichment of boundary objects used in CPD process.
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Keywords: post-merger integration; duality of integration; digital platforms; boundary spanning

Introduction

Mergers and acquisitions (M&A) help organizations improve their market share rather quickly and act as a key mechanism to realize synergies in size, geography, assets, people, or competencies and thereby create value (Epstein, 2005). Organizational integration during M&A involves the integration of strategic, operational, cultural, structural, and technological aspects. The ubiquitous nature of IT-driven business processes and the need for visibility of real-time information within and across boundaries drive the demand for an integrated common IT platform in merged organizations (Grudén & Strannegard, 2003). An integrated common IT platform not only improves an organizations' overall process integration capability, but it also positively influences firm performance (Rai *et al*, 2006). Acquirer organizations with strong IT integration capabilities and flexible IT infrastructure have demonstrated significant returns in the short term and greater operating performance and value over the long term (Tanriverdi & Uysal, 2011; Sarrazin & West, 2011). Despite the positive impacts of having a common IT platform in merged organizations, their development remains a significant hurdle in achieving the intended benefits of M&A (Chang *et al*, 2002) (Harrell & Higgins, 2002; Larsen, 2005; Huang & Chuang, 2007; Alaranta & Henningsson, 2008). For example, since business processes need to be restructured to enable effective common platform

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development (CPD) (Larsen, 2005), the relationship between restructuring business processes and CPD needs to be carefully evaluated. The nature and diversity of the processes that are integrated greatly influence the effectiveness of platform development process (Alaranta & Henningsson, 2007; Chao & Lin, 2009; Halonen *et al*, 2009). Knowledge of organizational and cultural context of the to-be-integrated processes (e.g., systems, processes, resources, clientele, regulations, etc.) becomes critical in balancing the need to preserve the distinctive characteristics of individual units with the need to provide a coherent platform for common operations (Wijnhoven *et al*, 2006; Mehta & Hirschheim, 2007; Jaakkola *et al*, 2010). The criticality of achieving such a balance and the difficulty involved in the process require that both business and IT stakeholders perform due diligence synergistically (Sarrazin & West, 2011).

Common platforms in which variability can be controlled can make context-sensitive adaptation more manageable while simultaneously balancing the need for stability and flexibility of business processes (Tilson *et al*, 2010). Despite the increasing importance of platforms, IS research has been less attentive to social and infrastructural issues in their development and use (Tilson *et al*, 2010). Common platforms allow organizations to let individual groups customize and co-create applications/services to meet their local needs while maintaining control over common behaviors in application or service ecologies through the creation of core services (Tilson *et al*, 2010). Common platforms become even more important in congeneric mergers which involve firms with interrelated, but not identical, business lines (Wen *et al*, 2005) and are driven by synergies in distribution channels, production technology, or customer base. With congeneric mergers, since complete dissolution of processes from the acquired firm is often not desirable, achieving stability through a common platform and maintaining flexibility for individual groups to customize the platform are critical. There has been little research done to understand how acquiring organizations in congeneric mergers establish such a balance. This gap in the literature motivates our research question: *How do contextual forces shape the process of balancing stability and flexibility in the development of a common platform in post-merger environment?*

Our paper is organized as follows: The literature on CPD in software engineering, which has tackled similar problems in developing software product families, provides theoretical background on how control points are exercised to support business process families that result from M&A. Details of our research methodology, research site, data collection and analysis are presented in the section after that. This is followed by the presentation of our research framework and a discussion on its validity in the subsequent section. We then discuss theoretical implications of our findings in light of extant research in the penultimate section and provide concluding remarks in the final section.

Theoretical background

We draw upon the extant literature on the role of IS integration in M&A, CPD, and boundary spanning to inform our empirical study.

Role of IS integration in M&A

IS/T integration encompasses technical and organizational integration. Technical integration covers integration of IS applications, IT infrastructure, IT work processes, workflows, and so on while organizational integration includes workforce and cultural integration of the IT organizations (Chao & Lin, 2009). Technical integration can be realized in many different ways (Johnston & Yetton, 1996; Harrell & Higgins, 2002; Brunetto, 2006) including (1) maintaining *status quo* (no integration of IS infrastructure), (2) using one company's infrastructure over the others, (3) using the best of breed, (4) acquiring new systems and migrating all the systems to this new system, and (5) outsourcing. We observed the following main themes in our review of literature on post-merger IS/T integration (summarized in Online Appendix A): (1) Decision models and frameworks to support and evaluate IS integration, (2) Factors that affect the effectiveness of or critical success factors in IS integration, (3) Governance modes, (4) Role of enterprise architecture, (5) Financial value realization, and (6) Role of IS architecture.

Although the current IS literature on M&A exposes us to various issues faced by organizations in the integration of IS/T infrastructure, there is paucity of work (with some possible exceptions of Dudas & Tobisson (2007) and Houwelingen (2008) on understanding integration activities. Specifically, there is limited understanding of the factors that enable or hinder the post-merger development of a common platform, which is perhaps among the most critical yet difficult tasks in post-merger integration. A few researchers have examined the role of enterprise architectures in enabling effective planning and implementation of IS-enabled development of a common platform in organizations (Dudas & Tobisson, 2007; Houwelingen, 2008). The use of macro and micro templates that document IS/T infrastructure analysis and decision tradeoffs can aid requirements determination and also contribute to knowledge sharing, shared decision making, and the establishment of mutual commitment between IS and business management (Main & Short, 1989).

Common platform development (CPD)

CPD approach has been utilized widely in manufacturing and software development industries (Meyer & Lehnerd, 1997; Northrop, 2002) to deliver customized products that meet individual needs while minimizing the cost of development. It strikes a balance between cost and marketability by allowing organizations to provide tremendous variety and individual customization at prices comparable to standardized products and services. As illustrated in Figure 1, it is done through large-scale planned systematic reuse of a common platform across a variety of products

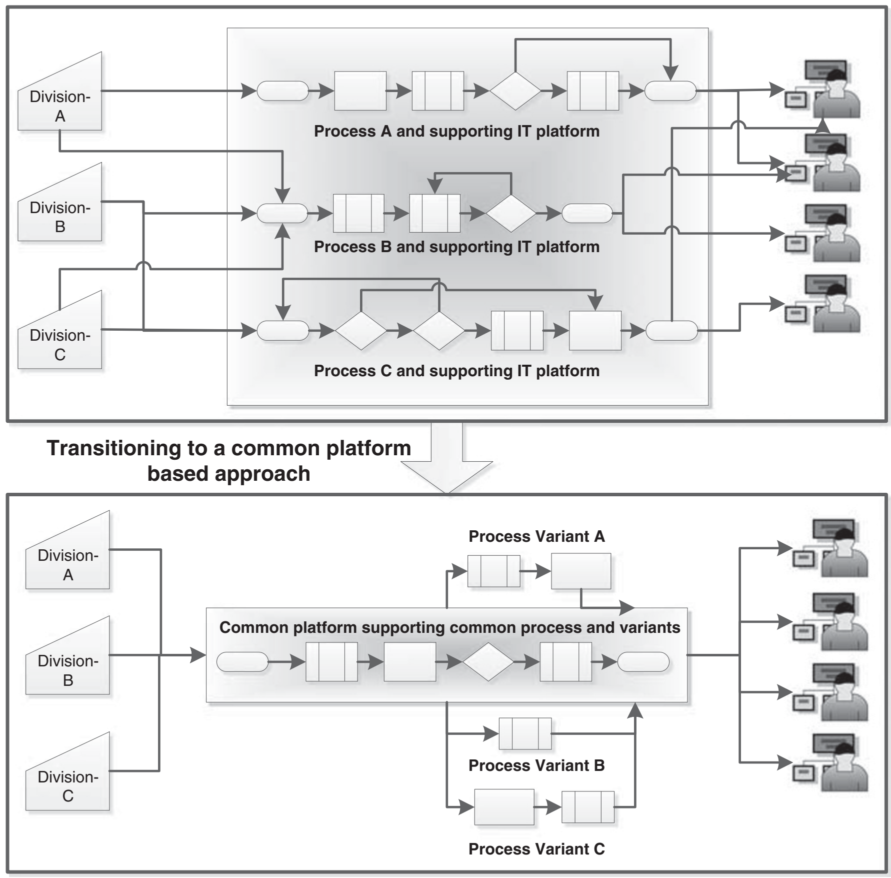


Figure 1 Transitioning to a common platform approach in post-merger environment.

and services (Keepence & Mannion, 1999). Introducing a common platform based approach requires extensive changes and close collaboration at various organizational, process, and technological levels (Kircher *et al*, 2006).

A primary goal in CPD is to understand and control common and distinguishing characteristics of similar systems through a systematic evaluation of these systems. This is done through variability management, which is similar to maintaining version control over time in conventional software development. However, in CPD the focus is on feature variations that occur over space as well as time, that is, across multiple similar systems and over time (McGregor *et al*, 2002). Variation points help to identify how and why product variants differ (Pohl *et al*, 2005). Platform-based approaches have recently gained wide recognition in modeling families of business processes as well (Reichert & Weber, 2012; La Rosa *et al*, 2013). However, much of literature on variability management is centered on modeling notations. This literature has not paid adequate attention to social aspects related to how process variations are identified, negotiated, and agreed upon.

Variability management, wherein common and distinct characteristics are identified, is a tedious and knowledge intensive activity because the knowledge about what variations are required and why, as well as how, they may be incorporated in product variants is scattered across different parts of the organization (Jiao & Tseng, 2000a, b). Further, it requires stakeholders to cross various functional boundaries and integrate knowledge fragments about variation points. The lack of a clear understanding of how to incorporate access points in a common platform can lead to expensive redesign. Variation points become focal in creating controlled access points in common platforms (Tilson *et al*, 2010).

There has been growing recognition in IS literature about platform-centric ecosystems (Tilson *et al*, 2010; Tiwana *et al*, 2010; Yoo *et al*, 2010; Yoo *et al*, 2012) and their capacity to enable flexibility (i.e., change) without sacrificing the pursuit of common core functionalities (i.e., control). This literature highlights the strategies for managing the tension between change and control (Tilson *et al*, 2010). These strategies are increasingly evident in various digital ecologies such as browsers and their add-on extensions, mobile platforms such as Apple's iOS and Google's Android, social media platforms, and gaming consoles (Tiwana *et al*, 2010; Tilson *et al*, 2012). The stability of a platform is essential in the enrollment of new development entities that generate new functionalities through applications, yet the platform also needs to be flexible enough to enable this generativity. Drawing development entities to exploit this generativity requires extensive boundary spanning inside and outside of organizations' boundaries. Despite the innovative ability to manage this tension that is increasingly common in practice, empirical research on how organizations transition toward common IT platforms to enable and exploit generativity in post-merger environments is scant. Also, the CPD approach has not been examined in a post-merger environment for creating and managing a family of similar business processes. We draw upon this literature to highlight how organizations establish a balance between stable and flexible aspects of a common platform.

Boundary spanning

Organizations represent a complex network of interdependencies among numerous specialized entities that need to cross their boundaries to work together, collaborate, innovate, or identify new opportunities. This ability is especially important in organizations that have evolved through M&A. Boundary spanning activities play a critical role in the development of a common platform that integrates business processes across multiple business divisions created as a result of M&A. The effectiveness of boundary spanning is driven by various factors including the nature of boundary spanning individuals and the nature of boundary objects (Pawlowski & Robey, 2004).

Boundary spanners play a crucial role in enacting change or innovation initiatives that cross multiple boundaries. Boundary spanners perform various roles such as ambassadors (who engage in political activities such as lobbying for support and resources, and buffering a group from outside pressure), task-coordinators (who facilitate information and feedback seeking activities), and scouts (who engage in environmental scanning activities) (Ancona & Caldwell, 1992; Yan & Louis, 1999). Sawyer *et al* (2008) observed five boundary spanning roles of IS development teams on the basis of their knowledge search and transfer activities viz., borrowers, isolationists, insulators, politicians, and exhibitionists that led to variations in ISD performance. Key activities performed by boundary spanners include searching, translating, and disseminating knowledge (Merminod & Rowe, 2012), knowledge acquisition (Kirsch & Haney, 2006), negotiation (Kirsch & Haney, 2006), consensus building, and conflict resolution (Carlile, 2002, 2004). While knowledge transfer focuses on sharing of explicit, codifiable knowledge, knowledge translation deals with sharing of contextual or socially constructed knowledge. Merminod & Rowe (2012) examine whether boundary spanners' role remains vital in post-implementation phase and how it evolves during this phase. They examine knowledge sharing effectiveness pre- and post-implementation of a PLM system in an international group with locations in China and France. Their findings suggest that the role of boundary spanners (outsourcing engineers in this context) becomes more critical in knowledge translation tasks to resolve and unify interpretive differences by creating shared meanings since common lexicon rules remained inadequate in enabling such translation (Merminod & Rowe, 2012). These boundary spanners achieved this by reducing cultural misunderstanding while also using advanced technological features such as 3D viewers and direct access to broad range of information to identify semantic gaps and improve the relationships with the suppliers.

Several factors affect the effectiveness of boundary spanning individuals. Boundary spanners rely on the knowledge of their organization's political context and the motivations of others to bring about change (Balogun *et al*, 2005). Developing a careful understanding of organizational culture and policies, awareness and adjustment of organizational processes according to cultural expectations, and skillful use of synchronous and asynchronous ICT and communication channels improves boundary spanners' effectiveness (Pauleen & Yoong, 2001). For individuals to emerge as boundary spanners (Levina & Vaast, 2005) they should be (1) capable of exchanging accumulated capital (economic, social, symbolic, or cultural) from one entity to another, (2) seen as legitimate or empowered (Kohli & Kettinger, 2004), and (3) motivated (intrinsically or extrinsically).

Drawing from the literature on situated learning in communities of practice, Pawlowski & Robey (2004) show how boundary spanning individuals, formal or informal (Johri, 2008), link know-how, know-why, and know-who from different domains (Fernandez & Gould, 1994). While past research has examined how boundary spanners connect with different communities and enable communication, in this research we examine how they help shape the process of post-merger CPD by enabling the processes of knowledge translation.

Research methodology

We use a case study research methodology to identify various contextual elements that shape the development of a common platform in a post-merger scenario. We examine how stakeholders from different business divisions establish a common understanding of their processes to inform the development of a common platform underlying a family of similar business processes.

Research site

ForestCo, a Fortune 100 company, is in the forest, paper and packaging industry with over 50,000 employees worldwide and hundreds of manufacturing and distribution facilities throughout North America. From its inception, ForestCo has aggressively grown through several acquisitions that had created four large business divisions. Although these divisions operated in the same industry, they represent diversified products and customer markets. ForestCo sells some products as raw material or partially finished components to other commercial businesses, wholesales some products for resale, and sells another group of products to retailers under its consumer brand.

To achieve efficiencies in transportation planning and related IT services across all business divisions, the top management team (TMT) of ForestCo decided to develop a common platform. Initiatives to develop a platform were launched three times, albeit unsuccessfully. The fourth initiative (which is the focus of our study) lasted nearly 9 months. It was deemed to be successful by the process stakeholders and the TMT. Our study investigates this post-acquisition platform development process.

The selection of ForestCo as the study site was driven by purposeful, theoretical sampling (Glaser & Strauss, 1967; Yin, 2009) – that is, the potential to investigate the post-merger development of a common platform across multiple business divisions. Theoretical sampling can help improve the external validity of the case study research (Eisenhardt, 1989). Our selection of ForestCo and the focal initiative were driven by following factors: (1) The site offered a unique setting to understand the process of post-merger development of a common platform. It offered a theoretically relevant organizational context because of its long history of growth through acquisitions as well as the difficulties it had faced in assimilating the acquired organizations into a single entity. (2) It also offered opportunities for disconfirming our expectations that the organization will be able to successfully develop a common platform because it had failed in at least three previous attempts at doing so (Markus, 1989; Dube & Pare, 2003). (3) Finally, despite the past failures, it had been actively trying to integrate diverse systems across its multiple divisions that came with the M&A. These criteria ensured that our selection of case study site provided a rich context to understand and develop insights on the development of common platform. Our objective was not to select a representative organization from which to generalize.

ForestCo's multiple divisions were vertically siloed, following the federated organizational governance structure that valued independence of the business divisions over which the corporate headquarters had minimal control. There was very little interaction, information sharing, or coordination across the business divisions. ForestCo had performed an analysis of the IT infrastructure used for transportation planning, and it highlighted the wide diversity among the systems used by the divisions. There were only a few systems that were common across divisions, but even these systems were implemented in each division as separate instances and were independently managed. As a consequence, when transportation service providers worked with multiple divisions, they had to interact with each division independently. This isolated nature of business divisions, diversity in their IT infrastructure, prior failed attempts at CPD, and the presence of federated governance presented a sufficiently rich context to study the development of a common platform.

Data collection

We used various data collection strategies (Lee, 1989) to ensure continuous engagement with the research site and collect rich qualitative data as summarized in Table 1.

Table 1 Data collection strategy

Data collection source	*Data collection details*
Primary sources of data	• Observation of eight all-day process integration workshops conducted over 4 months plus several workshop-related meetings involving members of CPO team that were conducted before, during, and after the workshop series • Brief informal unstructured interviews with workshop participants
Secondary sources of data	• Annual and quarterly reports, process documentations that were shared before, during, and after the workshops, and articles in trade publications

ForestCo conducted eight all-day workshops involving 10–15 senior executives that included the vice presidents and IT directors (who we identify as process stakeholders) responsible for managing transportation planning processes of the individual divisions. ForestCo's CIO and the initiative's sponsor attended some of the workshops. The Corporate Program Office (CPO) team managed the logistics of the workshops. Table 2 summarizes the various stakeholders that were involved in the integration workshops and the number of interactions we had with each group of stakeholders. This includes the interactions that occurred during the workshops, informal unstructured interviews, and meetings related to the integration workshops.

Process stakeholders used the time between the workshops to develop process models representing their transportation planning processes. In addition to the workshop meetings, the CPO team held several meetings with process stakeholders from each division individually before, during, and after the workshop series to handle issues such as pending concerns from previous workshops, administrative issues, techniques for process modeling, and evaluate progress made in the workshops.

We participated as observers in all the eight workshops and in six other meetings. Workshop meetings were audio-recorded with the participants' consent, yielding an extremely rich set of data of over more than 50 h. Additional informal interviews with process participants were conducted to seek clarifications on issues raised during the workshops. See Online Appendix B for an excerpt of the interview questionnaire. These interviews allowed us to explore the expectations of the participants. Detailed notes were also taken during each of the data collection sessions. Workshops and interviews were our primary source of qualitative data. Secondary sources of data included the process models developed by the process stakeholders and summary notes taken by the CPO facilitator, which were shared with us, as well as annual and quarterly reports and articles in the trade publications.

Table 2 Stakeholders participating in the workshops

Stakeholders		*Number of informants*	*Number of interactions with each group of stakeholders*[a]
CPO team		6	85
Senior management team		2	6
Process Stakeholders from all operational divisions	IT process stakeholders: IT directors	5–8	40
	Business process stakeholders: Executive vice presidents of transportation planning processes	5–7	40

[a]Includes workshop sessions, informal interviews, and other meetings outside the workshop setting.

Data analysis

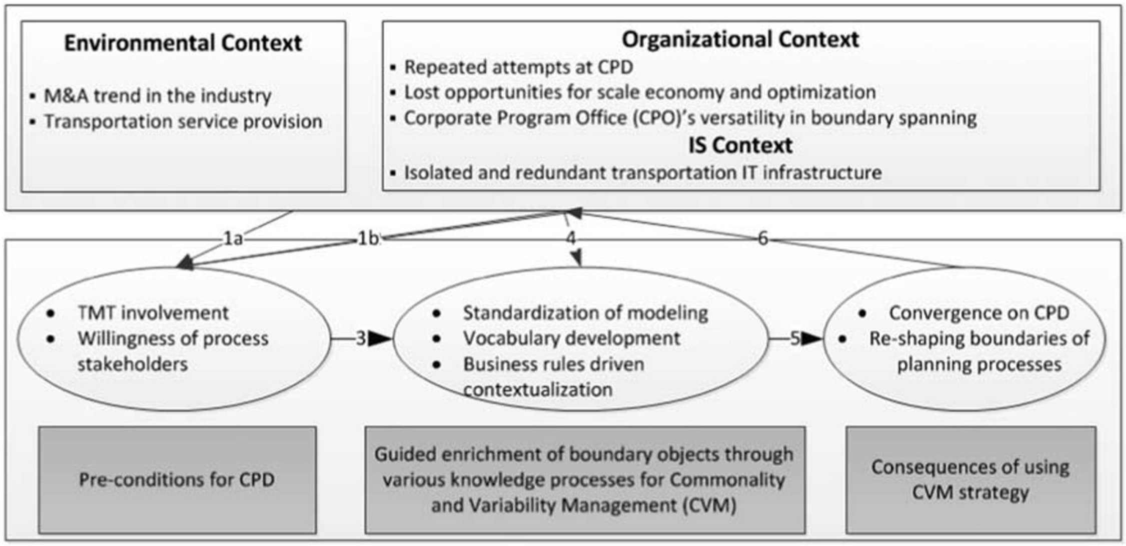
The data analysis focused on identifying key concepts, categories and relationships. The concepts identified in the section 'Theoretical background' – viz., context (environmental and organizational), boundary spanner roles (ambassadors, scouts, and coordinators), components of a platform (commonality and variability) – served as the seed concepts for initial coding of the data. New data were used to refine and/or modify prior concepts. We also actively searched for any new concepts and relationships that emerged from the data. While both researchers were actively engaged in data collection, one of the researchers did the preliminary data analysis and then the analysis results were debated until the researchers reached agreement through discussion and referencing back to the transcripts. This project is part of a larger research project for which extensive data was collected. Data collection was tightly interwoven with data analysis. As interviews were conducted, the transcripts were analyzed. Data collection and coding were done till 'theoretical saturation' (Eisenhardt, 1989) was reached, when additional data did not add to the concepts and categories developed.

The data analysis helped discover categories, relationships, and patterns in the data that formed the basis of the framework presented in the next section. On the basis of similarities and differences among the concepts, concepts were revised and merged to better represent the underlying meaning. These concepts were further validated and/or clarified by data from subsequent data collection efforts. As Strauss & Corbin (1990) suggest, theoretical relevance of concepts was indicated by their repeated presence or notable absence when comparing incidents. Online Appendix B provides examples of open coding. The data analysis process described above helped us establish the plausibility and consistency of the concepts, subcategories, and relationships that emerged. It also helped us establish that our conceptual framework satisfies well-established criteria for the credibility and validity that are commonly used in qualitative research studies (Miles & Huberman, 1994; Pawlowski & Robey, 2004).

Iterative comparison and contrasting of the concepts in the framework that emerged during our data analysis, and cross-examination across informants from the four divisions allowed us to sharpen the constructs (Eisenhardt, 1989) and ensure that categories were well developed and represented multiple viewpoints expressed by informants (Eisenhardt, 1989). This ensured that accumulating evidence obtained from multiple sources converged on well-defined categories. In ForestCo, we collected data through multiple means: observations (workshops, CPO team meetings), informal interviews (TMT, CPO team, and process stakeholders), and secondary documents (process models, presentations, notes, and other artifacts generated during various meetings and workshops). We were able to compare and contrast each category across four divisions, allowing us to replicate (Yin, 2009) and thus, increase confidence in our findings. The presence of multiple investigators allowed us to systematically recognize, discuss, and debate different interpretations and improve our understanding of the challenges faced by the informants in identifying conditions and strategies that enabled CPD. We presented our framework to key informants and made minor adjustments to the terminology used in the framework as a result.

Findings

Our research framework (shown in Figure 2 below) that was developed based on the analysis of data summarizes the contextual environment and the process through which stakeholders achieved convergence on the development of the common platform. This figure highlights how various contextual elements (increasing M&A activity and pressure to cut transportation costs, prior failed attempts to develop a common platform) motivated TMT's extensive involvement in the initiative (Arrows 1a and 1b). CPO team played an instrumental role in the initiative through various boundary-spanning efforts (with roles such as ambassador, scout, and knowledge orchestrator) throughout the initiative (Arrow 2). These organizational and IS contextual factors helped gain initial participation from all business divisions. The TMT's flexibility with its goals for the initiative and the willingness of the empowered stakeholders from multiple divisions to engage in the initiative provided



Legend for arrows:

- 1a and 1b: Contextual factors motivate extent of TMT involvement
- 2: Organizational and IS context influences process stakeholders' intent to participate in the initiative
- 3: Conducive environment to identify enabling strategies for CPD
- 4: Organizational and system context influences strategies identified to enable CPD
- 5: Enabling strategies facilitated boundary re-shaping through the development of specification for common platform
- 6: Increasing convergence on CPD enabled greater willingness of the process stakeholders through potential replacement of older IT infrastructure

Figure 2 A research framework for CPD.

a favorable environment for discussions on the merits of a common platform (Arrow 3). The CPO team supported these discussions by successively developing several rich boundary objects for supporting knowledge processes of knowledge transfer and translation (Arrow 4). These enriched boundary objects helped stakeholders to identify commonalities in their planning processes as well as unique division-specific requirements. This process helped the stakeholders converge on the feasibility of a common platform (Arrow 5). Further, it allowed stakeholders to recognize the limitations of existing systems and discover how a new common platform would address their concerns, leading to their sustained commitment to the initiative (Arrow 6). The boundary objects also helped the stakeholders understand how the boundaries of their processes and the resources needed to manage them may be reshaped by a common platform.

We describe the framework in detail in the following subsections.

Environmental context

In this subsection we discuss environmental forces that were external to ForestCo but specific to the industry in which it operates. These forces represent the aggressive nature of the industry. These were to a large extent beyond ForestCo's control but yet affected its strategy and operations significantly. These propelled ForestCo to pursue M&A aggressively.

M&A in the industry Mergers and acquisitions are very frequent in the industry in which ForestCo operates, especially after the revision of the merger guidelines in which antitrust authorities adopted a friendlier position in the mid 1980s. Nearly, about 40% of the capacity expansions at existing firms in this industry were achieved through horizontal acquisitions and over 40% of the organizations operating in the United States were involved in at least one merger during a 15-year period (Draffan, 2011). Additional factors that motivate this trend in this industry include (Pesendorfer, 2003): (1) savings through more efficient reorganization of production with larger production runs (Miller, 2002), (2) efficient allocation of inputs, and (3) increased effectiveness in sales and distribution networks.

ForestCo itself had been pursuing rapid expansion through a flurry of M&A transactions. A review of ForestCo' s annual reports revealed that it had gone through at least 20 M&A transactions in the two decades preceding its first attempt at integrating the business processes of acquired entities. These acquisitions had left ForestCo with myriad of transportation planning processes and their siloed IT infrastructure, making it difficult to develop any synergies that it had envisioned from M&A.

> We act like 'bunch of' [repeats] different billion dollar companies and that 'we're not' [repeats] one company … As we started to look at our strategy, one of the first things that's very very obvious is we just don't even have the visibility in what's happening in the supply chain, because everybody has different systems …. the

> objective is that of one implementation of the software, serving all the business units, we're going to have common processes.

This lack of synergy in post-merger environment had led ForestCo's TMT to designate the development of a common platform as a strategic initiative. It sought to achieve the efficiencies that had motivated M&A transactions and to seamlessly manage its family of transportation planning processes (Arrow 1a).

Transportation service provision Since the industry in which ForestCo operates relies on bulky raw material, it incurs very high transportation costs. In fact, in its industry, transportation costs represent the single most important cost and account for nearly 20% of the total cost of production for a major product line for ForestCo. ForestCo, like many organizations in the industry, relies on trucking companies to provide transportation services rather than maintaining its own private fleet. These trucking companies range from mom-and-pop operations to very large, national transportation service providers. ForestCo has a variety of contracts with the trucking companies, ranging from one-off contracts to long-term partnerships with these providers. At the time of the focal initiative, ForestCo was spending close to $1.5 billion on transportation. Therefore, TMT expected that even simple improvements in transportation planning would result in significant savings (Arrow 1a).

Organizational context

In this subsection we discuss forces that were internal to ForestCo that had been instrumental in reviving and effectively completing the current attempt at the CPD initiative.

Repeated attempts at CPD After the flurry of M&A transactions, ForestCo was organized into four business divisions, which were almost autonomously managed. Functions such as manufacturing and supply chain management were handled individually at each division, leading to significant replication in process functionality at all levels. A decade before our focal study, ForestCo embarked upon the first attempt at a CPD initiative. The main guiding principle was 'OneForestCo OnePlatform', which succinctly captured the fragmented and diversified nature of planning processes and the TMT's desire to act as one organization rather than as multiple organizations when dealing with transportation service providers. Although the TMT had realized the importance of having a common platform, it had done little to support, encourage, and empower the relevant process stakeholders. Process stakeholders were expected to attend the day-long meetings, perform process analyses and modeling activities, in addition to carrying out their daily responsibilities. As a result, the first attempt failed to gain sufficient participation.

> Okay I mean the thing was just in the fail from the beginning because it was not aligned. There was no executive support and plus resources were an issue. And we all had day jobs and it was the matter of who's going to crank up the analysis and who's going to help develop the strategy …

Three years after this first attempt, ForestCo rejuvenated its interest and organized a logistics roundtable to identify various opportunities in transportation planning. Around this time, one of the divisions decided to forgo participation in this integration attempt because it was heavily engaged with integrating a consumer products company that it had just acquired. This division and its acquired unit represented a large part of ForestCo that did not participate in the initiative. As a consequence, the stakeholders who did participate were unable realize the potential savings, create a shared understanding of process integration, identify appropriate targets, and set boundaries. It was also characterized by the lack of organized resources, leading to another failed attempt at CPD.

> … and again you know there were resource issues, we had great plans, great ideas and you know we set immense targets and said let's try to meet the targets and because of the resources it was not very successful …

A couple of years later, ForestCo tried again, albeit unsuccessfully. This third attempt, like its predecessors, failed to take off due to unrealistic targets and inadequate resources.

> And that's [after] another couple of years … for the first time we had all the business units representatives and we had executive support, but resources again was the big issue and like the previous initiatives we set huge targets and then we're trying to find ways to support those targets …

These repeated attempts underline the significance that ForestCo's TMT placed on developing a common platform that could simultaneously support the common and distinct aspects of transportation planning processes of the various divisions (Arrow 1b). Recognizing the strategic importance of this opportunity to pursue CPD, ForestCo established a CPO team that was tasked to identify solutions to the problems faced in the prior attempts.

Lost opportunities for scale economy and optimization Since each division was focused on optimizing its own specific transportation capacity requirements, optimization across divisions did not receive much consideration. This led to higher transportation costs. There were several occasions where transportation planners lacked visibility into information on incoming trucks to-be unloaded at their facility. These trucks left the facility empty when in fact they could have been filled with finished goods on their outbound journey.

> I mean, just the visibility aspect of this can be huge for us. Right now [truck is] going from the mill to a plant. I mean the plant doesn't even know that the truck is coming in. You know, he [transportation planner] working on one side of [plant], he's worried about

getting everything else, he might be calling for trucks over there and well over here, the two trucks sitting back there, waiting to get unloaded.

In the absence of an integrated planning process and a common platform to support it, transportation service providers were getting transportation requirements separately from each division. There were vast variations in the requirements of these divisions that made it difficult for providers to consolidate and optimize their own internal allocation of transportation services and as a result they passed on higher transportation costs to ForestCo. The new platform was expected to bring visibility to operations across divisions on a day-to-day basis, and to help discover new opportunities for collaboration across business divisions. This would, in turn, allow for more predictable demand for transportation services and better contract terms with service providers and reduce the transportation costs (Arrow 2).

> One of the reason, we keep preaching all over the place about the collaboration and trying to get everyone in the world to jump in the bandwagon [is] that the carriers because they have risk of the running the truck empty, if you marry up enough freight and put up enough partners together, they don't have that risk anymore, the price goes down and then, then contractual rates are no brainer at that point. That's why we keep going after people.

Without the common platform, however, such visibility was difficult to achieve and most business divisions were concerned about whether and how their unique transportation requirements would be supported in the common platform. All the prior attempts had failed to show that such integration of processes and the development of common platform were even feasible. CPO team worked persuasively to overcome these concerns and took on multiple roles to facilitate the discussions.

CPO team's versatility in boundary spanning The role of the CPO team evolved significantly over the series of workshops. They worked vigorously behind the scenes to keep the workshop series going. As the process stakeholders began to take a stake in the initiative, the CPO team took on various roles from being ambassadors to scouts to orchestrators to facilitate the process stakeholders at different stages of the deliberations (Arrow 2).

CPO team as ambassadors Since its formation, CPO team took several steps in establishing CPD as a strategic initiative within ForestCo. The team analyzed the issues that had plagued the prior failed attempts. It identified stakeholders within each division who were empowered to negotiate on the functionalities that needed to be incorporated in the platform and approve final decisions. It began communicating with these stakeholders even before the workshops began to explore their concerns and gain their participation in the workshops. CPO team then shared these concerns with the TMT to explore the range of possible actions to address them that included subsidizing the cost of CPD development and maintenance as well as running a pilot to demonstrate the feasibility and potential value from CPD. The outcomes of these approaches began to mold process stakeholders' attitudes toward the common platform. CPO team also began championing the initiative and documented the costs of not having such a common platform.

> … There is no opportunity across [ForestCo] to really leverage our collaborative bases and it is also costing us a lot of money to do what we do now. There is a huge payback for [ForestCo].

As the workshops began, there were also discussions about the scope of the initiative in terms of different transportation modes and the global nature of the processes. CPO team, working with the TMT and division stakeholders, carefully pared down the scope of the initiative. It encouraged the process stakeholders to evaluate the operational flexibilities that may be gained by adopting a common platform. For example, individual divisions would be shielded from operational disruptions that may result from IS and IT upgrades, have access to best-of-breed solutions for transportation planning, and make changes to their planning processes with increased agility. It also succeeded in brokering a deal with a vendor to accommodate distinct features requested by the divisions in the development of the common platform. This deal significantly eased stakeholders' concerns about the CPD.

These various actions of CPO team which were ambassadorial in nature – opening up communication channels between TMT and stakeholders, persuading and molding stakeholders' views toward CPD, and negotiating with TMT, stakeholders, and vendors – helped ForestCo to overcome several issues that impeded past attempts, helped recognize the CPD as a feasible strategic initiative to all involved parties and engage them in identifying common and distinct aspects of their processes.

CPO team as scouts Once the CPO team brought the concerned process stakeholders for negotiations at the workshops, team members also acted as scouts looking for mechanisms to facilitate a shared understanding of the core planning process. When stakeholders brought process models in multiple formats and at different levels of granularity, CPO team researched various modeling notations that would be expressive yet simple. After a careful review, CPO team finally developed a swim-lane based process template that could be used to share details of individual planning processes. The team offered assistance to individual divisions in their modeling efforts. It also scouted for industry best practices and frameworks to guide the stakeholders' efforts in mapping their processes. This led them to identify and adapt the industry standard SCOR model for ForestCo's business environment.

> [CPO] team also looked at how the Truckload freight process could be defined using standard process steps

[of SCOR] and activities as a way of developing the "OneForestCo" process flowchart.

Process stakeholders were immediately able to relate many of their transportation planning tasks to the SCOR model and identify areas where they deviated from the industry practices, for better or worse. Thus the scouting activities to identify the SCOR model and the swim lane template enabled stakeholders to identify the common thread across their planning processes. It also eased stakeholders into documenting their planning processes consistently in their CPD endeavor (Arrow 4).

CPO team as knowledge orchestrators As the workshop participants began developing a clear grasp of the features and functionalities of the common platform, the CPO team became responsible for coordinating the various activities. It worked to shield process stakeholders from laborious process analysis tasks by developing analytical resources in the form of boundary objects such as ontology-based vocabulary (discussed in the section 'Vocabulary development') and context-based process templates to document business rules driving decisions in the planning function (discussed in the section 'Business rules driven contextualization'). By deploying these resources, boundary spanners enabled and empowered the process stakeholders to translate their division-specific knowledge of transportation planning such that stakeholders from other divisions could relate to it. Such translation helped stakeholders develop an understanding of and appreciate the diverse business and cultural backgrounds of other divisions before their acquisition by ForestCo. It also enabled the identification of division-specific requirements that had to be maintained in the common platform (Arrow 4).

CPO team ensured that stakeholders provided rigorous business cases when requesting custom development to support distinct aspects of their divisions. The deliberations also highlighted a need for an enterprise wide IT governance group to evaluate and approve custom developments in the future. Again, CPO team worked closely with each division to identify individuals to serve on this governance group. As stakeholders continued their deliberations, CPO team meticulously documented the discussions with details on decisions made, best-in-class practices to be shared across all divisions, and the features to be implemented in the common platform. In this role, CPO team performed various activities that enabled knowledge documentation, sharing, and vetting.

Table 3 summarizes impact of various boundary-spanning activities that CPO team performed during the CPD process.

IS context

Isolated and redundant transportation IT infrastructure Since ForestCo had grown rapidly through a series of acquisitions over the past two decades, it had a proliferation of systems to support various transportation-planning activities. In some cases, even within a single division, multiple systems for performing the same task were used in different plants. Many of these systems were aging and becoming ineffective at handling higher volumes of loads. Also these systems worked in isolation, making it difficult to perform route optimization. Many transportation service providers, especially smaller ones, were moving toward web-based EDI. Maintenance of these systems including dedicated EDI systems used by individual divisions was becoming increasingly expensive due to the demands placed on IT personnel and the duplication of resources:

> There are some shocking number on IT hardware and maintenance reduction of about 800k and then on the labor side about a million dollar including IT and business mainly due to the duplicate processes.

In later workshops, as stakeholders had become increasingly convinced of the value of CPD, the consolidation of the various systems was also recognized by process stakeholders as critical for increased operational efficiencies and cost savings. Discussions on how to address these pain points and their desire for new functionalities led stakeholders to slowly view the CPD initiative as a solution (Arrow 2):

> As for my wish list I want to integrate [carrier management system] and off course I would like to eliminate manual input if possible … Another pain point is, for us, is [not] having a central place for all the data. Lack of central place is a big pain. And that's because we track and want to gather information on lot of transaction that happen from the time order hits the system until it is booked or shipped. There is lot of information in our transactional environment, as well as that we pull from the [freight management system] and shipment history. But they are all over the place. And a central data warehouse will be helpful.

Thus, environmental, organizational, and IS related contextual elements influenced the nature and the extent of the TMT's involvement as well as the process stakeholders' desire to participate and contribute toward CPD.

Pre-conditions for CPD

Willingness of process stakeholders The CPO team brought together process stakeholders from all business divisions to participate in a series of workshops. These were senior executives directly responsible for transportation planning of their respective divisions, heads of IT organizations that supported transportation planning, and staff members involved in transportation planning. Unlike the prior initiatives, these process stakeholders were empowered to make decisions on behalf of their divisions.

> There is one more thing I want to emphasize is. It is of empowerment. You guys are empowered. you know we all have struggled to [define] what empowerment means … there is a gray area.. but you were selected

Table 3 Mapping of boundary spanning activities to their impact on CPD process

Post-merger CPD activities	*Boundary spanning activities*	*Impact of boundary spanning on CPD*
Establishing CPD for transportation planning as a strategic initiative	Championing: Ambassadorial	Gaining support from TMT
Scrutinizing past failures to understand how to communicate with various business divisions	Analyzing to reach out: Scouting	Improved understanding of what clouded stakeholders' vision for common platform development
Communicating about strategic nature of CPD with relevant stakeholder groups from all business divisions	Opening up communication channels and informing: Ambassadorial	Improving willingness of stakeholders to participate in CPD
Using various techniques to gain support from different business divisions in CPD workshops	Persuading and molding: Ambassadorial	Improving willingness of stakeholders to participate in CPD
Negotiating with stakeholders and TMT to establish realistic scope	Negotiating: Ambassadorial	Improving willingness of stakeholders to participate in CPD
Negotiating with vendors to enable custom development for supporting distinct features (variation points)	Negotiating: Ambassadorial	Development support for enabling common and distinct features (variation points) in the platform
Cataloging and analyzing strategic utility of myriad systems across all divisions	Gathering information and analysis: Scouting	Helped to identify what systems would become part of common platform and what new features to build
Working with outside stakeholders to identify ways to document and share process knowledge	Gathering information and resources: Scouting	Improved ability of stakeholders to identify common and distinct features of their planning process
Documenting and adapting industry best practice frameworks for mapping transpiration planning tasks	Gathering information and resources: Scouting	Standardization in documenting current process activities to enable discovery of common and distinct features of transportation planning
Analyzing process models and documenting transportation terminology that were not interpreted uniformly across multiple divisions	Enabling inter-division process knowledge translation: Orchestrator	Common ontology-based vocabulary to reduce confusion and chaos in identifying common and distinct features
Enabling stakeholders to identify, develop, and share knowledge of variation points for their processes and what, why, and how behind them	Enabling inter-division process knowledge translation: Orchestrator	Identification of distinct features at the individual division level
Coordinating and Integrating knowledge of commonalities and variation points using business rules through a cost-benefit analysis across divisions	Harmonizing analysis efforts across multiple divisions through effective knowledge translation: Orchestrator	Instrumental to advance consensus on the implementation of common platform
Coordinating with various divisions to form enterprise-wide IT governance team for CPD initiative	Coordinating: Orchestrator	To rationalize, prioritize, and approve common and division-specific features for CPD inclusion going forward

> because of your knowledge, power, and respect across the organizations and ability for making decisions.

IT personnel accompanying these senior executives worked to ensure that the CPD would indeed support their needs. This immediate support of IT personnel during the workshops helped to accelerate decision-making. As the initiative progressed, process stakeholders began to respond positively to the various ideas discussed in the workshops. Specifically, they identified exceptions or variations to the common platform that would be necessary to satisfy their unique needs (Arrow 3). The process stakeholders were candid in explaining not only how their concerns may be addressed through CPD but also were equally candid about how compromises resulting from the CPD would affect their planning practices. This allowed the TMT to take corrective steps as needed (discussed below), further bolstering the confidence of the process stakeholders.

TMT involvement Intense M&A actions and the desire to contain the cost of transportation service provision through common platform had led the TMT to be actively involved in addressing concerns of business divisions. Through the CPO team, they ensured that most issues plaguing the past initiatives were adequately addressed. TMT also took a much more realistic and malleable approach in the focal initiative to avoid the pitfalls of the previous attempts. With the CPO team facilitating the initiative, TMT worked with individual divisions to ensure their participation in the workshops. It offered to compensate divisions for the development and maintenance costs of the common platform. To convince stakeholders of the feasibility and potential savings from CPD, it launched a pilot project involving Division s1 and 3. This pilot also demonstrated that while there may be overall gains for the entire organization occasionally some divisions may incur losses. However, TMT guaranteed that such losses will be absorbed at the corporate level and that the individual divisions will not be penalized.

> No one is going to get shot here. Let us just go out there, do the analysis, do the development.

When the agenda for the initiative was initially developed, it was envisioned that process stakeholders would consider all modes of transportation planning (truck, rail, and intermodal). During the initial workshop, however, stakeholders signaled that they possessed very little expertise on some of these transportation modes and the inclusion of additional participants to consider them would excessively enlarge the scope of the imitative.

> You know we talked about this [rail based transportation] earlier, you know this is kind of like a whole, whole different group that's also driving this … but I think it's, a pretty big chunk of what we are trying to do in a very aggressive timeframe as well … We need a whole session for it, Existing transportation representatives would need to be supplemented with additional subject matter experts on Rail.

TMT accordingly narrowed the scope of the initiative to only truck transportation. As another example of flexibility, TMT worked with business divisions to find an acceptable solution in limiting the international scope of the initiative when stakeholders from Division 4 wanted to include international processes. To secure the participation of Division 4, TMT intervened and negotiated to include international transportation only to and from Canada and Mexico. This compromise tentatively addressed the needs of the Division 4 without increasing the scope of initiative excessively.

With the participation of process stakeholders and the TMT's willingness to address their concerns, the CPD initiative moved to the next critical step of actually identifying common and unique aspects of their planning processes (Arrow 3).

Commonality and variability management

We now discuss various strategies that CPO team implemented in its effort to facilitate the identification of common planning processes across all the divisions as well as business exceptions uniquely distinct to each division. Each strategy incrementally enabled process stakeholders to translate transportation planning knowledge, which they had deemed to be unique to their division, and recognize the feasibility of developing a common platform and reach agreement on CPD (Arrow 5).

Standardization of process modeling efforts Each of the business divisions outlined the nature of their transportation planning processes in the first workshop. However, each division had created process models at different levels of granularity and had used different modeling notations to represent their processes. For example, one division had created straw-man models using Microsoft PowerPoint that depicted the entire planning process in just one slide whereas another had created detailed step-by-step illustration of transportation planning processes using flowcharts that spanned over multiple pages. These differences had made the identification of common tasks, systems, data flows, and so on across divisions for transportation planning impossible.

> What we reviewed today with business divisions shows the wide variation in detail … we need to talk about what is the template, what is the standardized format that we're going to use to [model your processes]. Put them all in the same format, so we all are talking about same [level].. We'll need to identify the [common] functions.

To facilitate discussions, it was clear that process models needed to be developed using a common modeling formalism. To standardize the modeling efforts, the CPO team developed a template for process documentation based on the activity diagrams using the 'swim lane' format. Despite this, in the next workshop there were still wide variations in the granularity in the process models of various divisions.

Stakeholders felt that if they went any deeper there would be nothing in common:

> I want to note that, if we go to too deep a level of detail, I think ability to get a match across all these pieces is going to deteriorate rather rapidly. High level of process flows is as far as this is concerned is what we need to go with. I don't know if there is an expectation by [CIO] that every keystroke that my people do is same as every keystroke that your people do.

To address this concern, the CPO team suggested that an industry standard SCOR model be used to identify the key processes in transportation planning and to provide guidance on the level of detail with which process models may be developed. This standardization allowed process stakeholders to develop an initial set of models that were specified with similar levels of detail, enabling smoother knowledge transfer. These models helped create a shared understanding of how each division performed transportation planning and how it differed across divisions.

Vocabulary development While the use of SCOR models defined the scope of the modeling effort, process stakeholders were still unable to understand each other's transportation planning processes. During the 3rd workshop, the CPO team realized that this difficulty stemmed from the divergent use of transportation planning terminology across divisions to refer to similar planning tasks. Different divisions used different terms to refer to the same concept or worse till they used the same term to refer to different ways of performing similar transportation planning tasks. The process of identifying common tasks within transportation planning became very chaotic. For example, most of ForestCo's divisions used the terms 'order' and 'load' interchangeably; however, Division 1 used these terms very differently. In Division 1, the term 'order' refers only to an 'order' that came from the production division whereas the term 'load' refers to multiple 'orders' that were put together to create a truckload. In contrast, the other business divisions usually handled only truckload (TL) orders rather than less than truckload (LTL) orders and used the terms 'order' and 'load' interchangeably. Similarly, in Division 3, MAX rate indicates the highest possible rate paid to carriers for a given load and is determined on a load-to-load basis.

> Through the order processing, we are given a MAX rate and we can pay up to that rate. And if we go over that rate it goes back to our order people to find out from where they are going to get extra money from, whether it's going to come from the [plants] or from the generators.

In contrast, Division 1 sets up a dummy carrier called MAX in their system for every route whose rate is the average rate for that route to ensure that its internal optimization system will always find the lowest rate carrier.

> So what happens is this: the system will never plan for a carrier over MAX. However you've got those carriers in there, so if you're in a pinch, and you got nothing, you can always go over your average, but somebody has to do that, the system is going to never do that for you. And if you think about that, what that system does, is it always puts us below average. So MAX rate is a huge part of what we do to, to communicate dynamically to the planners what is the cost that we're willing to pay in a particular lane.

These examples highlight how the lack of a common vocabulary across divisions that were pieced together through several years of M&A created barriers in developing the shared understanding that is needed to achieve CPD, thereby creating barriers for knowledge sharing. As the process stakeholders continued to reconcile the diversity in the terminology used, they considered whether they had to select a set of terms that can be used uniformly across all the business divisions. After realizing the complex and time consuming nature of using this option, they decided that it would be more useful to simply agree on the definitions of the terms and creating a mapping across them. It was recognized that while it is virtually impossible to mandate the use of the same terminology to refer to similar process concepts, it is possible to develop a shared understanding of the terminology used by different divisions and create a systematic mapping among them.

> … I'm thinking one of the things that we want to do is [use] some kind of standardization [in the terminology used]. Use of the vocabulary is one way to achieve the connection in our processes and avoid some of our confusion …

The workshop facilitator developed a vocabulary to document the terms used and developed a mapping among these terms to aid in the translation of the transportation planning knowledge across divisions to some extent. During the next workshop, process stakeholders debated, refined and finalized this vocabulary. This vocabulary was used in succeeding process development efforts. This vocabulary was deemed crucial in reducing confusion and improving understanding of diverse practices and disparate systems that were inherited through several acquisitions.

Business rules driven contextualization The standardization of process modeling and the use of a common vocabulary helped ForestCo stakeholders make some progress in developing a common basis without causing confusion. However, they continued to struggle in the task of identifying common planning tasks. One of the guiding principles 'No Sacred Cows' drove each business division to re-think and rationalize any unique aspects characterizing their planning processes. Process stakeholders faced difficulties in identifying justifications for processes that seemed unique to their divisions. These divisions had continued to follow the same processes they had used before being acquired by ForestCo. This led the CPO team to develop an elaborate template to document contextual elements underlying the planning processes in terms of business rules, decisions, alternatives, assumptions, inputs, outputs, exceptions, systems used, and process inefficiencies.

> Well, [if] we look at our business assumptions behind these things, then we might be able to find out why we do what we do so differently …

Using this template, Division 2 stakeholders took the initiative to add these contextual details to their process models. The use of this contextual information allowed stakeholders to uncover and articulate various hidden assumptions, formal, and informal business rules behind their processes. The quotes below represent some of the differences in the business rules followed by Divisions 1 and 2.

> We [Division-2] let the carrier accept, reject, or counter a load offer … we keep track of those with EDI messages if he [carrier] is EDI-capable.

The above is in contrast with the load tendering process followed by Division 1. In Division 1, once the carrier assignment is done through optimization, the load is directly tendered to the carrier. If the carrier cannot accept it for some reason, it contacts Division 1 representatives, a practice known as 'tender-with-confidence'. All the stakeholders deemed this business rule as one of the best business practices and agreed to incorporate it in the common platform to improve the overall efficiency and effectiveness of load tendering. Another example related to the verification of insurance compliance. Since Division 2 dealt with a wide range of carriers (from large national carriers to small local mom-and-pop carriers) on a regular basis, it always verified that a carrier had the required insurance liability coverage before any load was assigned to it. Even though verification of insurance compliance was an enterprise-wide policy, other business divisions did not verify coverage as they almost always used large national carriers who had more than adequate coverage. This was deemed as a 'best practice' and was included in the common platform.

Teasing out such contextual details allowed process stakeholders to separate actual planning tasks from their business rules and thus enabling a seamless knowledge translation process. The use of business rules made it easier for them to not only identify best practices to be followed across all divisions but also justify unique practices for their business divisions.

Convergence on CPD

On the basis of the initial understanding of the requirements for the common and variable elements of the platform, CPO team conducted a pilot project involving a part of Divisions 1 and 3's planning processes. Contextualized process models that captured business rules helped in the identification of common and distinct elements. Division 1 had implemented an older version of a system that would be part of the common platform and provided a live demonstration of planning functionalities during the last two workshops. This demo along with the contextualized process models made it easier for the process stakeholders to envision the use of common platform for their planning tasks. These discussions, in turn, helped stakeholders to coalesce around a shared understanding of the common and variable features that should be included in subsequent versions of the common platform (Arrow 6).

> I mean we have all realized that we all do very similar things and the [common platform] would be able to help us achieve higher optimization with the volume. I think we all realize the value in doing this by now. I think we are already building a pretty good case for the value and what is common. In terms of discussing common, I mean we all book a load, ship a load, and tender a load. I mean we are pretty much common across these areas already. But I think what we ought to look at is the areas that we don't have.

The use of various techniques to manage common and variable aspects of the transportation planning processes of all divisions resulted in the redefinition of multiple planning processes into a common process and a platform to support it. CPO team established an enterprise-wide Transportation IT Governance group (ITG) to manage the monumental task of managing the requirements for the common platform. This group was responsible for providing cost-benefit analysis for the functionalities requested by the divisions and how those functionalities would impact the overall operations. It would also work with the stakeholders to collect additional transportation systems related requirements. CPO team identified members for the ITG with the help of stakeholders who had participated in the workshops.

> ITG governance group will be led by transportation representatives from each business unit, to provide checks and balances, to review and approve changes to IT systems and business processes.

This group would be responsible for championing any new technological innovations proposed by the process stakeholders to the TMT. Business process stakeholders from individual divisions would provide information on the impact of not having a feature or functionality on their operations.

Establishing validity of the research framework

To ensure the credibility and the validity of our case study, we address following concerns: (1) theoretical sampling (discussed in research methodology), (2) logical coherence, (3) theoretical relevance, and (4) relative explanatory power (Glaser, 1978; Eisenhardt, 1989; Hedman & Kalling, 2003).

Our research framework identifies the mechanisms involved in CPD and various factors that influence this process as well as interrelationships among them. In addition, our research framework provides a process perspective on the enabling conditions for post-merger integration at the system level. For example, our findings suggest that CPD initiatives require intense engagement of top management as well as empowered stakeholders who fully understand the implications of a common platform on their business operations. Since the knowledge that is necessary for successful CPD is dispersed across diverse divisions, knowledge transfer, translation and transformation need to be supported through the use of rich boundary objects. Our model also highlights that the CPD process is iterative in that stakeholders' commitment to the common platform and the functionalities that are identified for inclusion in the common platform are emergent, and are shaped by boundary spanning practices. In addition, our framework also highlights the dynamic interaction between the environment and the CPD process. Thus, our study provides a comprehensive understanding of how various contextual elements and organizational mechanisms influence CPD.

We established the theoretical relevance of various constructs in our framework by ensuring their repeated presence or notable absence (Strauss & Corbin, 1990) in the data. Specifically, Glaser (1978) suggests that theoretical relevance can be established by not only generating new categories but also by integrating extant categories that could possibly fit from the literature in novel ways on an emergent basis. During the coding process, we followed both strategies: new categories were generated (e.g., CPO team versatility, TMT role and stakeholder empowerment, strategies to enable CVM) and extant categories from boundary spanning literature (such as boundary spanner roles, boundary objects) were integrated into our coding process in an emergent fashion, leading to the development of a coherent research framework that is presented in Figure 2. While findings from such analyses may be particularistic, we use these findings to develop a general explanation by following well established procedures used in such studies (Eisenhardt, 1989; Orlikowski, 1993) through analytic generalization (Yin, 2009). For example, while at ForestCo, the strategy to identify and manage commonality and variability involved successive enrichment of boundary objects; in other M&A contexts such a CVM strategy may take a different form. Furthermore, to enhance the internal validity and improve generalizability of the research (Glaser, 1978; Eisenhardt, 1989), we relate our findings to the literature on the role of IS integration in M&A, digital platforms, and boundary spanning in the section 'Discussion'.

Extant research on the role of IS integration in M&A has been relatively silent on post-merger integration activities such as the processes used in the development of common platforms and critical issues involved in implementing these processes. Similarly though the notion of the required balance between stability and generativity has been addressed conceptually in the literature on digital platforms, our understanding of underlying strategies to enable such balance is limited. Our research framework addresses both of these concerns by identifying contextual factors at different levels (industry, organizational, and system) along with the enabling pre-conditions, strategies to enable CVM and the consequence of using it in the development of the common platform at ForestCo.

Discussion

The following research question guided our study: *How do contextual elements shape the process of balancing stability and flexibility in the development of a common platform in post-merger environment?* We answered this question by developing a systematic framework that explains how a CPD process is shaped. It also identifies boundary spanning roles, boundary object development strategies, and their influences on CPD process. Below, we elaborate on this framework by proposing the duality of process integration. We also note that this framework challenges the amethodical view of IS development that is often evident in the literature on CPD.

Enabling generativity of digital platforms in post-merger environment through boundary spanning

The notion of 'generativity' of digital infrastructures advanced by Tilson *et al* (2010) maps well to the characteristics of the platform that was under development at ForestCo. While Tilson *et al* (2010) advance this notion based primarily on their examination of information technologies that have reshaped societal and industry level use, our study offers insights into development of a digital infrastructure (i.e., the common platform) at the organizational level. By examining the role of individual stakeholders, groups and organizational structures (such as the CPO) in shaping the development of the common platform, our study addresses the call by Tilson *et al* (2010) to empirically examine the influence of paradoxes of change and paradoxes of control that uniquely distinguish digital infrastructures. Specifically, our study highlights how the TMT facilitated the development of a platform that supports stable and common functionalities, but also offers flexibility to individual divisions to manage their business processes that are unique to their setting. Our account of boundary spanning practices that helped to achieve this objective provides detailed descriptions of how the social and technical elements shape the development of the digital infrastructure. It shows how CPO team enabled stakeholders in the identification of architectural characteristics that would ultimately empower them to exploit the generativity of the platform. It also highlights the tensions that organizations encounter in establishing a delicate balance between the 'opposing logics of stability and flexibility' (Tilson *et al*, 2010). Further, it also examines the tension between exercising and loosening control that is essential for the development of the platform. Thus our study represents one of the first attempts at examining these phenomena at the organizational level.

While a large number of studies examine the factors that enable or hinder IS/T integration processes (Giacomazzi *et al*, 1997; Robbins & Stylianou, 1999; Dudas & Tobisson, 2007; Alaranta, 2008), there is limited attention paid to understanding how CPD may be achieved. Our study contributes to this literature by identifying specific mechanisms that have been used in the successful development of a common platform. Our study suggests that the enrichment of CPD efforts successively through standardization of process modeling, semantic clarification through vocabulary development and contextualization through business rules facilitates the development of a shared understanding of the core features that should be incorporated in the platform. Although prior studies (Alaranta & Kautz, 2007; Chao & Lin, 2009; Halonen *et al*, 2009) have suggested that environmental factors affect platform development, our study enhances this literature by identifying the challenges that result from these factors, which in turn are addressed by strategies used in the focal organization. Our findings also confirm some aspects of the dynamic system of IS integration in M&A situations that was developed by Henningsson & Carlsson (2011). For example, our study highlights how the potential of IS to provide synergy was initially not recognized by the majority of the stakeholders. Our study offers a nuanced view of the evolution of the viewpoints of these stakeholders in that the same stakeholders began to slowly recognize the synergies that could be achieved by IS integration. This evolution was facilitated by the boundary spanning activities carried out in the platform development process. Thus our study suggests an additional link between synergistic potential and intentions and reactions not included in the DySIIM model (Henningsson & Carlsson, 2011).

Our findings confirm the observations by Levina & Vaast (2005) on the characteristics of successful boundary spanners. Specifically, by the judicious use of economic and social capital, the CPO team helped foster a climate that was conducive to successful negotiations among the process stakeholders. Since the CPO team had the backing of senior management, its legitimacy was unquestioned (Kohli & Kettinger, 2004). The CPO team also gained legitimacy by their actions. Their move away from playing a central role to that of a support role also dramatically increased the confidence and trust placed in them by process stakeholders. Merminod & Rowe (2012) similarly observed that boundary spanners such as local outsourcing engineers can enable trust development by translating situated knowledge by using their expertise in bridging technical, organizational, and cultural boundaries. Our study reinforces the critical and irreplaceable role of the boundary spanners in enabling knowledge translation processes in CPD initiatives, as also emphasized by Merminod & Rowe (2012) in their investigation of boundary spanners' roles in PLM implementation. Thus, knowledge translation activities performed by boundary spanners can also enable the balancing act of achieving stability and generativity in CPD initiatives. Also, ability of boundary spanners to mobilize various strategies to enable the identification of common and variable elements (McEvily *et al*, 2003) helped them achieve shared understanding (Bechky, 2003) of the core processes that will be incorporated in the common platform.

Managing 'drift' in CPD

Our study offers interesting insights on the notion of 'drift' observed by Nandhakumar *et al* (2005) in a study of ERP implementation. The concept of drift enables us to understand how legacy practices that one is aiming to purge can embed themselves in new IT implementations (Wagner *et al*, 2011). It refers to the processes that mitigate the gap between open IT implementations and legacy practices (or the situated interventions of use). Organizational members' desire to include legacy practices may lead to unexpected deviations in the course of implementation. Drift processes aimed at including legacy practices may or may not lead to desirable outcomes depending on how they are managed (Elbanna, 2008). Given the scope of change and complexity of large-scale IT project implementations, drifts are inherent (Elbanna, 2008) and mechanisms and procedures for managing drifts must be identified (Mintzberg & Westley, 2001). This requires the presence of configurations that are capable of initiating, monitoring, and balancing diverse interactions among the stakeholders and various events that may trigger such drift processes (Lyytinen *et al*, 2009). These tasks require that individuals in such configurations are capable of spanning various boundaries and developing boundary objects that can initiate and balance interactions among the stakeholder groups to skillfully balance positive and negative impacts of the drift processes. At ForestCo, while intentionality and social structure indeed played a significant role, the reshaping of the views of the various stakeholders achieved through boundary spanning activities helped reduce resistance to the development of the common platform, and thus minimize drift. The development of common platform in post-merger environment is essentially an exercise in re-shaping the boundaries of internal divisions, an undertaking rife with conflicts and drifts, requiring the presence of smooth interfaces to mend fences (Hanseth & Braa, 2001). At ForestCo, these interfaces were provided in the form of boundary spanners and boundary objects. For example, activities of the CPO team helped the stakeholders better understand how the affordances provided by the common platform would re-shape boundaries of their planning processes and the unique variants that may be derived from it to meet the goals of individual divisions. While the boundary spanning activities of the CPO team helped minimize stakeholders' tendencies to introduce excessive number of division-specific features to protect their 'turf', the CPO team also encouraged stakeholders to identify any legacy practices that may in fact be essential in the common platform. This suggests that the CPO team in its role of boundary spanner was able to shape drift processes to the initiative's benefit. Thus, while our study recognizes the emergent nature of the trajectory of CPD, it also identifies how through the appropriate manipulation of contextual factors (such as organizational incentives provided by the TMT) and organizational action (through boundary spanning activities) the trajectory of an IT project can be controlled to some extent and manage drift in platform development.

Duality of process integration

Our study develops a framework that illustrates how a CPD process is shaped by environmental, organizational, and system context characterizing post M&A organizations. It demonstrates the duality of CPD process, a dynamic interaction between the environment and the CPD process. As the common platform was developed to suit the environmental and organizational context, it changed the context characterizing platform development. For example, as a result of the CPD, IT and other organizational resources were re-allocated, mitigating some of the challenges initially faced in achieving post-merger integration. The new environment re-shaped the development of the successive versions of the evolving common platform. In another example, as stakeholders used the CVM strategy to analyze common and variable requirements of their platform, they became aware of how common platform may in fact be more beneficial and became increasingly involved in the development of the platform. Thus, the common platform is not only shaped by environmental and organizational context but was also critical in re-shaping the context. Thus, the common platform which corresponds to a modifiable artifact through its existence (Orlikowski, 1992) also continually affects its context as it evolves. We refer to this reciprocal interaction between the common platform and the context of process integration, *a duality of process integration*, following the concept of duality of technology (Orlikowski, 1992; Orlikowski, 2000).

Our study shows that neither the common platform nor the context for process integration remains fixed due to this duality. The implication of this duality is that it is essential to continuously monitor and maintain the fit between the context and the evolving product platform when operating in dynamic contexts. The common platform is tuned and refined as its development unfolds, and the attendant needs for common and variable features of the platform evolve. In this sense, CPD is emergent during its development life-cycle. Adopting this view would increase the likelihood of success in CPD. Further, complementing the amethodical view of systems development (Truex *et al*, 2000), our study illustrates that the CPD and even the context of process integration are a product of constant social negotiation and consensus building. In other words, the common platform and the context in which it is developed are also emergent during development.

Conclusions

Contributions to theory

Our study contributes to the literature on CPD and IS integration in post-merger contexts by providing a systematic framework to help understand and guide CPD. By proposing the duality of process integration, the framework also calls into question the amethodical view of IS development that is often evident in the literature on CPD. The framework identifies boundary spanning roles, boundary object development strategies, and their influences on CPD. These findings offer fresh insights into the role of contextual factors that shape CPD beyond what is reported in the current literature. Our framework empirically demonstrates how the balance between stability and generativity in platform development may be achieved. We also discuss how drift in platform development may be managed through effective boundary spanning efforts.

Contributions to practice

Our study examined prior attempts at the development of a common platform that were unsuccessful. In contrast, the focal initiative was deemed successful. This contrast brings into sharp focus the critical role boundary spanning activities that helped the organization achieve success with the initiative and offers opportunities for us to develop prescriptions for practitioners engaged in CPD. Practitioners can draw from the conceptual framework (shown in Figure 2) to drive their efforts toward CPD in post-M&A environment. Our study suggests that both the development of the common platform as well as the organizational environment need to be flexible. IT organization and business stakeholders need to continuously monitor and adjust both common platform and the organizational strategy to guarantee success in CPD.

The findings identify the organizational and environmental factors that need to be considered, and the strategies that may be used to address the issues that arise in CPD. Specifically, the roles of the support personnel such as the CPO team and the strategies for identifying commonality and variability in the common platform are expected to be of significant interest to practitioners. Since an integrated IT platform strategy improves organizations' overall process integration capability and hence firm performance (Rai *et al*, 2006), the CPD approach is likely to ease the integration of IT infrastructure in future M&A waves. The boundary spanning practices observed in our study (summarized in Table 3) provide concrete activities that organizations can engage in when creating common platforms to achieve such process integration.

Limitations and future work

Generalizing our findings to other contexts that involve CPD should be done with caution as the findings may be particularistic to use of specific environmental, organizational, and IS context we studied. However, our study provides fertile ground for quantitative studies that can examine specific factors that help organizations in shaping a common platform. The link between the development of such a common platform and organizational performance is a subject of ongoing research. Future research can also focus on key differences between the specific contextual elements observed in our study and other types of contextual elements that have been observed to influence IS project success. Recent research (Sandberg *et al*, 2013) suggests that the platforms and organizational strategy need to coevolve to maintain an appropriate level of fit. The investigation of how such a fit can be achieved through organizational action (say, through boundary spanning practices) is an interesting avenue for future research.

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